

COL774 / COL7341

Oct 31, 2025

Unsupervised learning:- (x^i, y^i) i.i.d.



$\{x^i, y^i\}_{i=1}^m$
 $\hookrightarrow$ Test time: $x \rightarrow \hat{y}$
 $\hookrightarrow \hat{y} = y$

K-Means } Unsupervised learning Algorithms
 GMM
 EM
 PCA 3 Dimensionality Reduction
 learning Thing

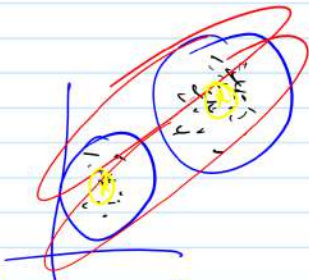
$\{x^i, y^i\}_{i=1}^m$

$p(x^i; \theta)$

K-Means:-

Clustering Algorithm:-

Data has a fixed # of clusters (categories)



$\{x^i\}_{i=1}^m$ $x^i \in \mathbb{R}^n$ K - # of clusters

Objective:-

- (A) Points within a cluster are close to each other
- (B) Points in different clusters are far apart

$C^k \in \{1, \dots, K\}$

$\mu^k \in \mathbb{R}^n$

$\hookrightarrow$ cluster centroids / cluster representatives

$J(C, \mu) =$

$\{C^k\}_{k=1}^K$ $\{\mu^k\}_{k=1}^K$

$\hookrightarrow \arg \min_{C, \mu} J(C, \mu)$

$$\frac{1}{m} \sum_{i=1}^m \|x^i - \mu_{C^i}\|^2$$

Mixed Optimization

Problem $\rightarrow$ Discrete & Cont. Parameters

(A) Assume: μ^k 's are given:-

$$\min_C J(C, \mu) = \min_C \frac{1}{m} \sum_{i=1}^m \|x^i - \mu_{C^i}\|^2 \rightarrow (1)$$

(Maximization step)

$\{C^k\}_{k=1}^m$

$$\equiv \min_{\{C^k\}_{k=1}^m} \frac{1}{m} \sum_{i=1}^m \|x^i - \mu_{C^i}\|^2$$

$C^{(1)}$: - pick the elements whose representative is closest to $x^{(i)}$

(B) Assume C 's are given -

$$\min_{\mu} J(C, \mu) = \min_{\mu} \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{C^{(i)}}\|^2 \quad (2)$$

E-step (Expectation step)

$$= \min_{\mu} \frac{1}{m} \sum_{i=1}^m \sum_{k=1}^K \mathbb{1}\{C^{(i)}=k\} \|x^{(i)} - \mu_k\|^2$$

$$\downarrow C^{(i)} = \frac{1}{m} \sum_{k=1}^K \sum_{i=1}^m \mathbb{1}\{C^{(i)}=k\} (x^{(i)} - \mu_k)^T (x^{(i)} - \mu_k) \quad \uparrow (x^{(i)} - \mu_k)^T (x^{(i)} - \mu_k)$$

Differentiate w.r.t μ_k :

$$\frac{\partial J(C, \mu)}{\partial \mu_k} = \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{C^{(i)}=k\} 2 (x^{(i)} - \mu_k) (-1)$$

$$\Rightarrow \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{C^{(i)}=k\} (x^{(i)} - \mu_k) = 0$$

$$\Rightarrow \mu_k = \frac{\sum_{i=1}^m x^{(i)} \cdot \mathbb{1}\{C^{(i)}=k\}}{\sum_{i=1}^m \mathbb{1}\{C^{(i)}=k\}}$$

K-Means:-

$t \leftarrow 0$
 $\mu \leftarrow \text{init}()$
 do
 $\{$
 $C^{(t+1)} = \arg \min_C J(C, \mu^{(t)})$
 $\mu^{(t+1)} = \arg \min_{\mu} J(C^{(t+1)}, \mu)$
 $t \leftarrow t+1$
 $\}$ (until !converge)